

```
from google.colab import files
uploaded = files.upload()

import cv2
import matplotlib.pyplot as plt

img = cv2.imread(list(uploaded.keys())[0])

img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

sobel_y = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)

sobel_y_abs = cv2.convertScaleAbs(sobel_y)

plt.figure(figsize=(12,5))

plt.subplot(1,3,1)
plt.title("Original Image")
plt.imshow(img_rgb)
plt.axis('off')

plt.subplot(1,3,2)
plt.title("Grayscale Image")
plt.imshow(gray, cmap='gray')
plt.axis('off')

plt.subplot(1,3,3)
plt.title("Sobel Y Edge Detection")
plt.imshow(sobel_y_abs, cmap='gray')
plt.axis('off')

plt.show()
```

No file chosen

Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

Saving logo080.png to logo080.png

Original Image



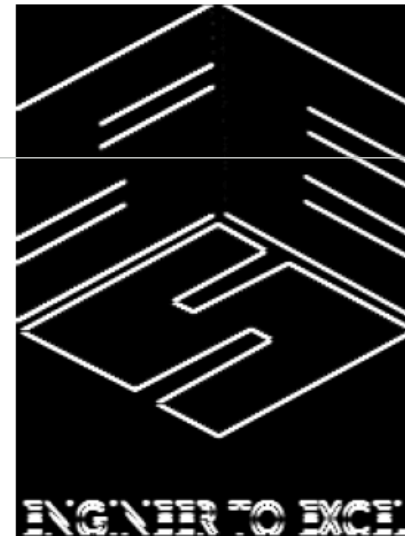
ENGINEER TO EXCEL

Grayscale Image



ENGINEER TO EXCEL

Sobel Y Edge Detection



ENGINEER TO EXCEL